pfi

Python for Fixed Income

COURSE 1 · PYTHON FOUNDATIONS

Visualization

WEEK 3 · DAY 1

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This course is educational content created in a personal capacity. Nothing here is investment advice or a recommendation to buy or sell any security. All portfolio data is synthetic.

A summary hides things. A chart shows them.

Last week ended with tables

Value by sector, the split by grade, the average coupon: each answer summarized 200 bonds in a line or two. An average spread says nothing about whether the bonds sit close to that average or far from it, and a table of 200 rows cannot be read at a glance.

200
bonds, in one table

7
kinds of chart

2
libraries

Every chart describes the data. None suggests a trade.

The portfolio manager has asked for pictures of the book

How are the spreads spread out?

Histogram · Box plot

How do they differ by rating?

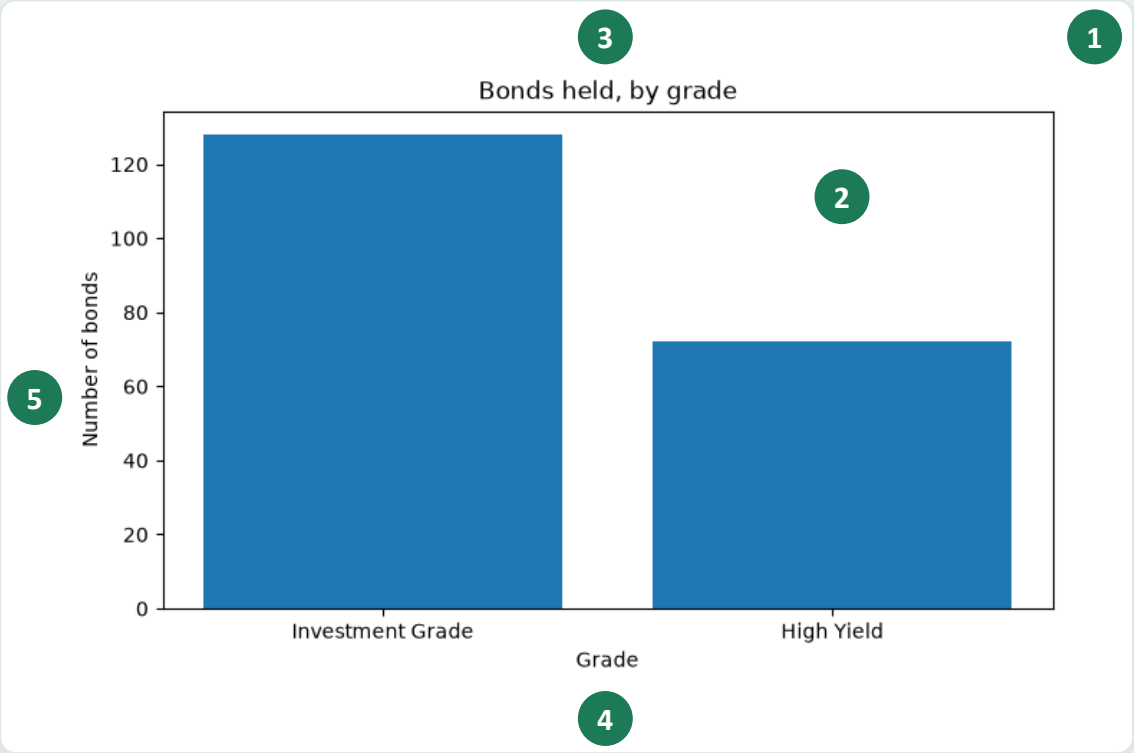
Bar chart · Line plot

Which measures move together?

Scatter plot · Pair plot · Heatmap

`matplotlib`	the base charting library of Python
`seaborn`	built on it: statistical charts in one line

The anatomy of a chart



Behind the bars: 128 Investment Grade, 72 High Yield

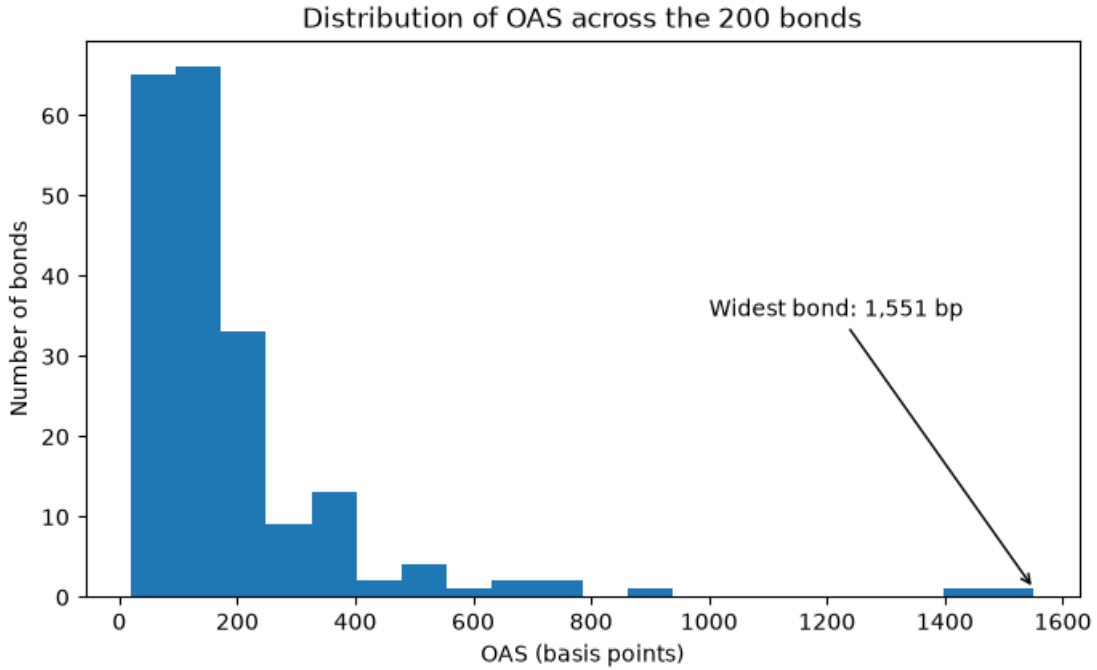
```
book['grade'].value_counts()  =COUNTIF(V:V, "High Yield")
```

Part	matplotlib	Excel
1 Whole picture	`fig, the figure`	Chart Area
2 The plot	`ax, the axes`	Plot Area
3 Title	`ax.set_title()`	Chart Title
4 x axis label	`ax.set_xlabel()`	Axis Titles
5 y axis label	`ax.set_ylabel()`	Axis Titles
Size	`figsize=(8, 4.5)`	drag a corner

Title and Axis Titles are under Chart Elements, the plus sign.

```
fig, ax = plt.subplots(figsize=(8, 4.5))   # make
ax.bar(grade_counts.index, grade_counts.values)   # draw
ax.set_title('Bonds held, by grade')   # label
ax.set_xlabel('Grade')
ax.set_ylabel('Number of bonds')
plt.show()   # show
```

Histogram: how one column is distributed




```
ax.hist(book['oas'], bins=20)
sns.histplot(data=book, x='oas', bins=20, ax=ax)
```

What to read off it

164
of 200 bonds below 250 bp

12
bonds above 500 bp

1,551
bp, the widest OAS

Read from the x axis up: each bar covers a range of OAS, and its height is the number of bonds in it. A crowd on the left and a long thin tail on the right: the column is skewed. With bins=5, one bar holds 173 bonds and the shape is hidden.

The same in Excel

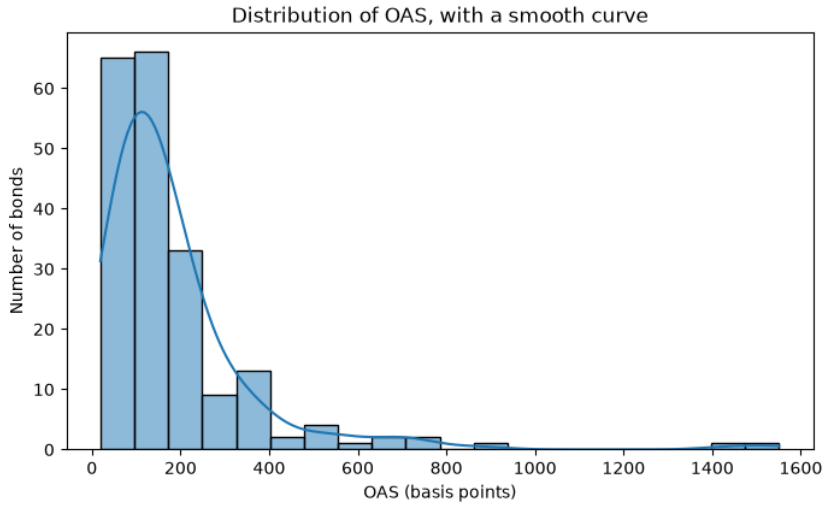
Excel 2016 and later

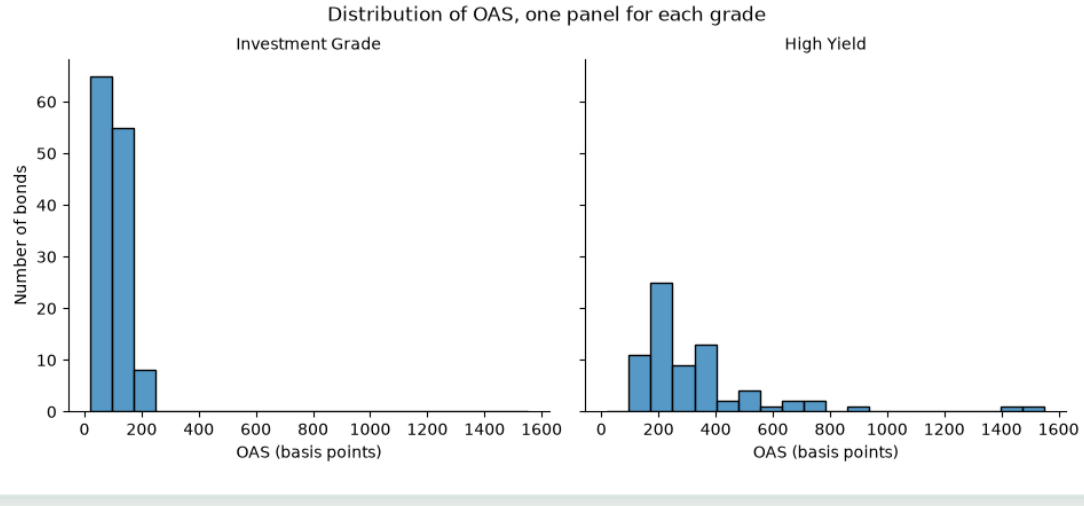
Insert, Insert Statistic Chart, Histogram

`=COUNTIFS(M:M, ">=19", M:M, "<325.4")` returns 173

Mean and median: =AVERAGE(M2:M201), =MEDIAN(M2:M201)

Histogram options: curve and panels

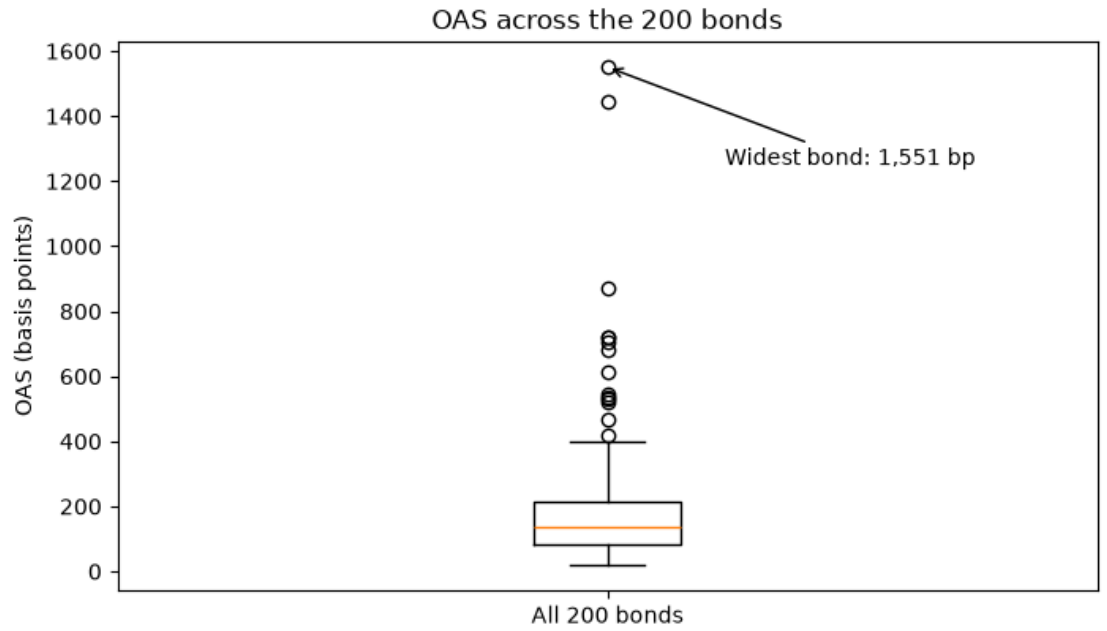





```
sns.histplot(data=book, x='oas', binwidth=50, ax=ax)
sns.histplot(data=book, x='oas', bins=20,
             kde=True, ax=ax)
sns.displot(data=book, x='oas', col='grade')
```

binwidth= sets how wide each bar is, and bins= how many bars. kde=True adds a smooth curve: read the bars for counts and the curve for shape. col= draws one panel for each group, on the same scale: Investment Grade is a compact hill, and High Yield has the right tail.

Box plot: five numbers in one shape




```
ax.boxplot(book['oas'], tick_labels=['All 200 bonds'])
# a list of columns draws one box for each
```

What to read off it, from the top of the chart down

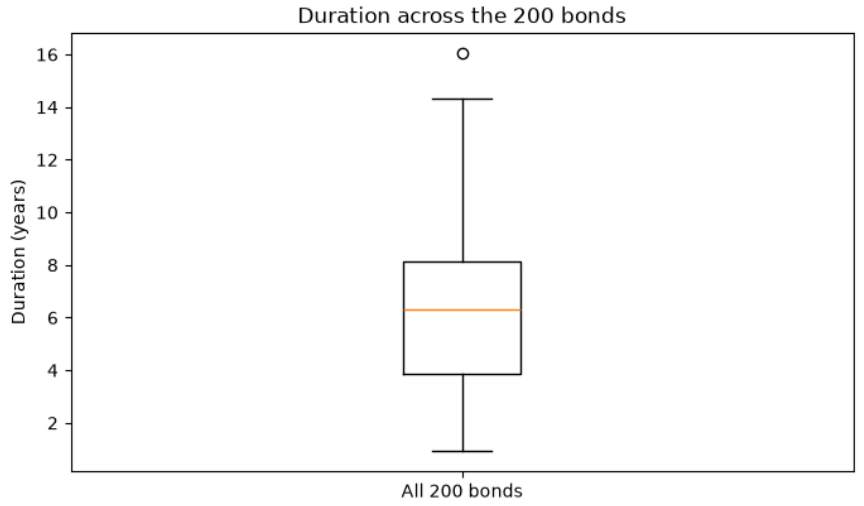
Part of the shape	OAS, in basis points
Points beyond the whisker	14 bonds, 418 to 1,551
Upper whisker ends	401 (the limit is 409.875)
Top of the box: third quartile	213
Line in the box: median	135.5
Bottom of the box: first quartile	81.75
Lower whisker ends	19, the lowest OAS

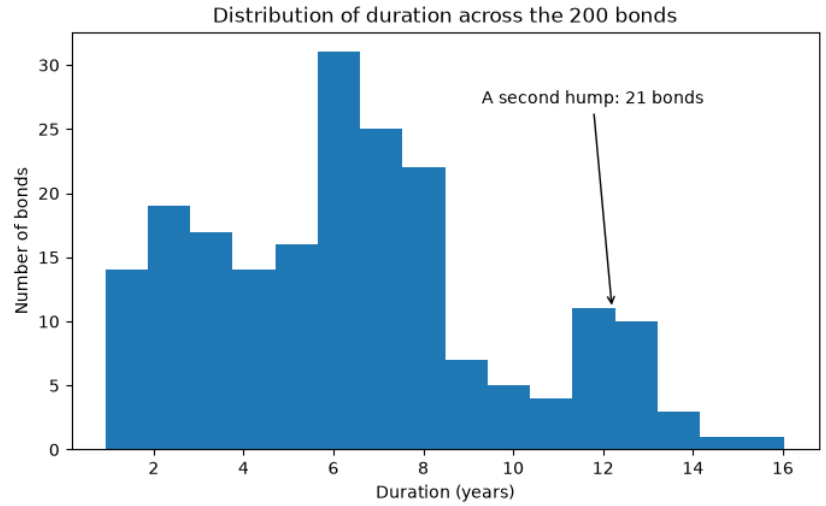
A whisker stops at the furthest bond within 1.5 x IQR of the box. Side by side, by grade: the middle half is 66 to 129.75 bp for Investment Grade and 205.25 to 368 bp for High Yield. The two boxes do not overlap.

Excel: Insert, Insert Statistic Chart, Box and Whisker

`=QUARTILE.INC(M2:M201, 1)` returns 81.75. With 3: 213

What a box plot hides





Two humps often mean two kinds of bond in one column. Draw the histogram before you trust a box.

6.31

years, the median: the box says nothing is odd

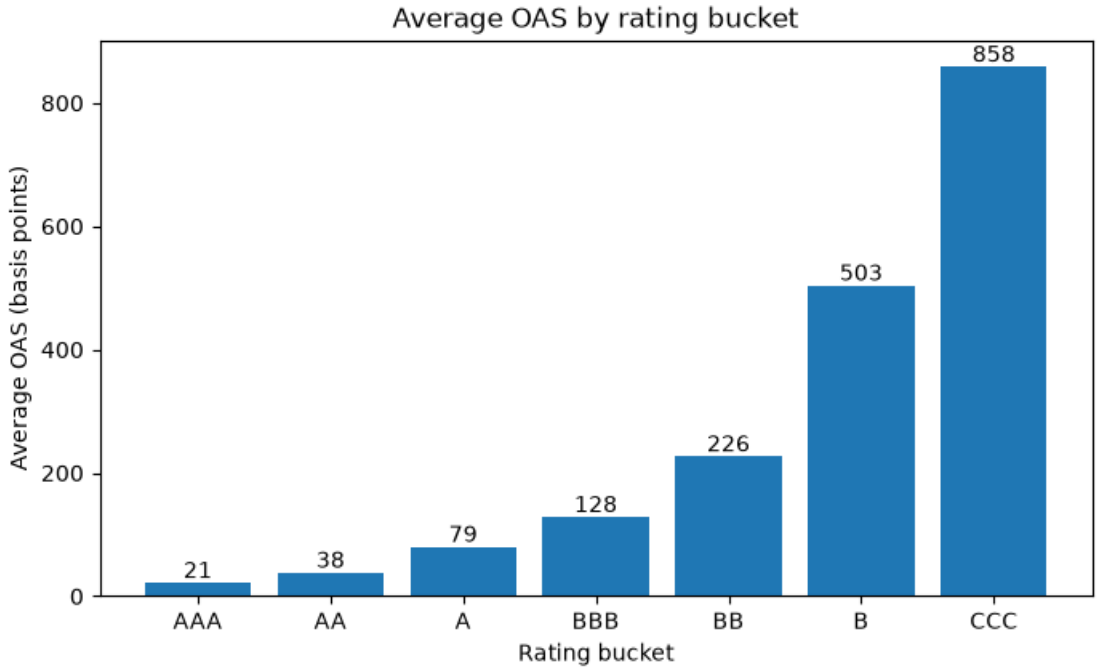
3.88 to 8.14

years, the middle half of the bonds

21

bonds in the second hump, 11.3 to 13.2 years

Bar chart: one number for each category




```
bars = ax.bar(mean_by_bucket.index, mean_by_bucket.values)
ax.bar_label(bars, fmt='%.0f')
```

What to read off it

21

bp, the average for AAA

127.82

bp, the average for BBB

858

bp, the average for CCC

Each bar is taller than the one to its left, and the steps grow: about 48 bp from A to BBB, about 355 from B to CCC. A bar shows one number per group. The B bar says 503, and the median B bond is at 409.5.

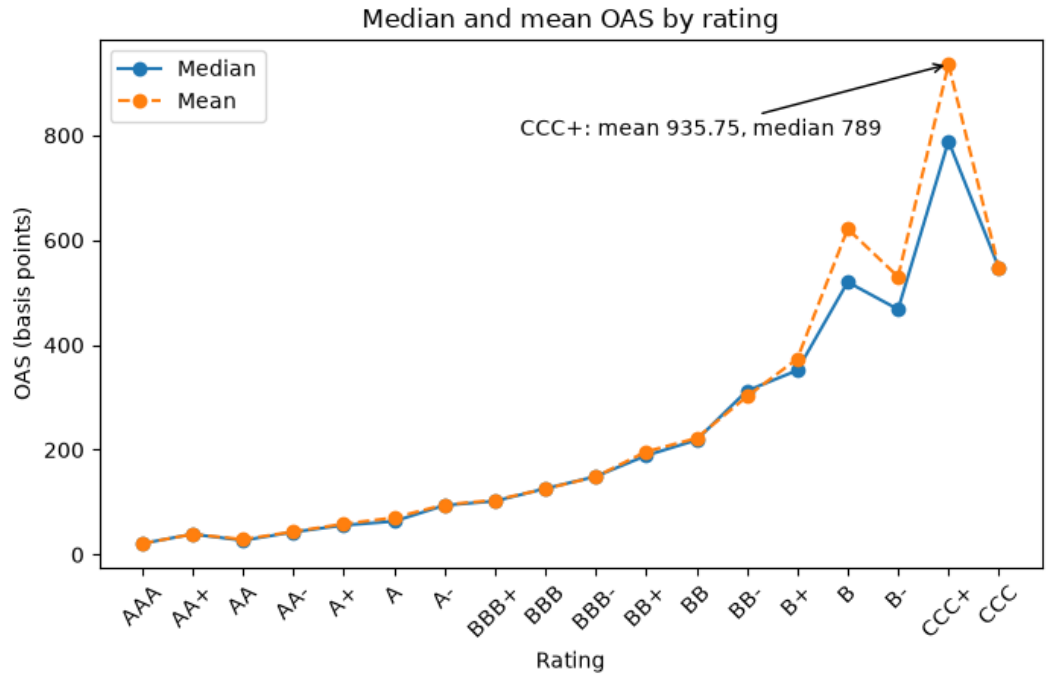
The same in Excel

Insert, Insert Column or Bar Chart, Clustered Column

`=AVERAGEIF(U:U, "BBB", M:M)` returns 127.82

Pivot table: bucket in Rows, Average of oas in Values

Line plot: a number along an ordered axis




```
ax.plot(by_score.index, by_score['median'], marker='o')
# a second ax.plot() draws a second line
```

What to read off it

21

bp, the median at AAA

149

bp, the median at BBB-

789

bp, the median at CCC+

The line rises slowly to BBB-, then steeply. It dips where a rating has one bond: AA+ and CCC. The mean and the median part at B (621.6 against 520) and at CCC+ (935.75 against 789), where the two widest bonds sit.

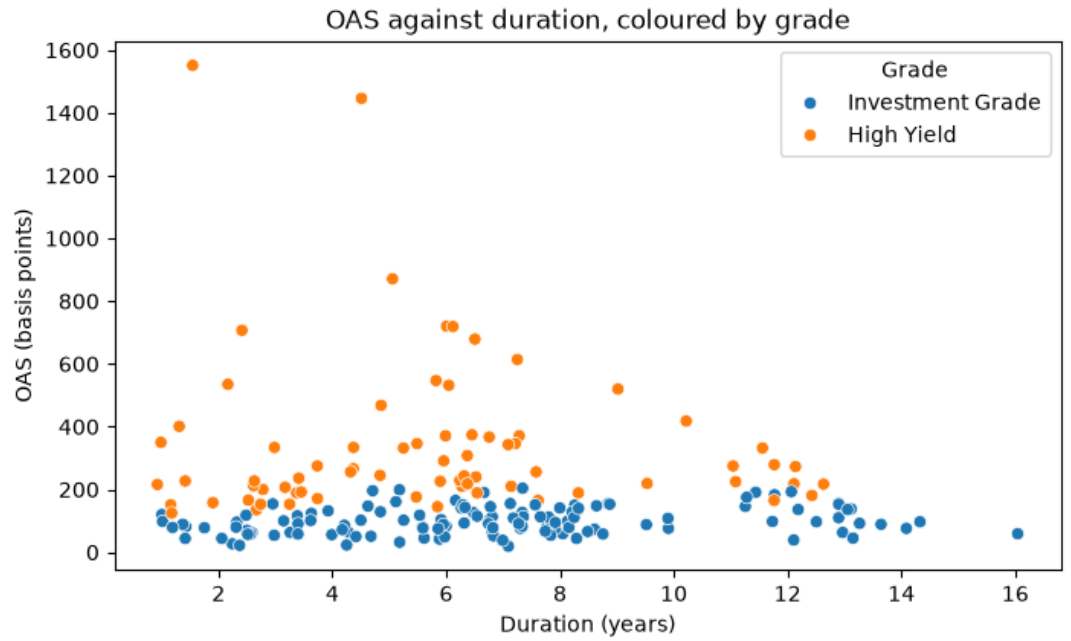
The same in Excel

Insert, Insert Line or Area Chart, Line with Markers

`=MEDIAN(IF(T2:T201=10, M2:M201))` returns 149

A second selected column becomes a second series

Scatter plot: one point for each bond




```
ax.scatter(book['duration'], book['oas'])
sns.scatterplot(data=book, x='duration', y='oas',
                hue='grade', ax=ax)
```

What to read off it

0.99
correlation of OAS and yield

-0.11
duration and OAS, whole book

0.24
within Investment Grade

Nothing is averaged: all 200 bonds are on the chart. Coloured by grade, the cloud is two layers. Investment Grade is a flat band with no bond above 204 bp. High Yield runs from 125 to 1,551. A correlation does not say one column causes the other.

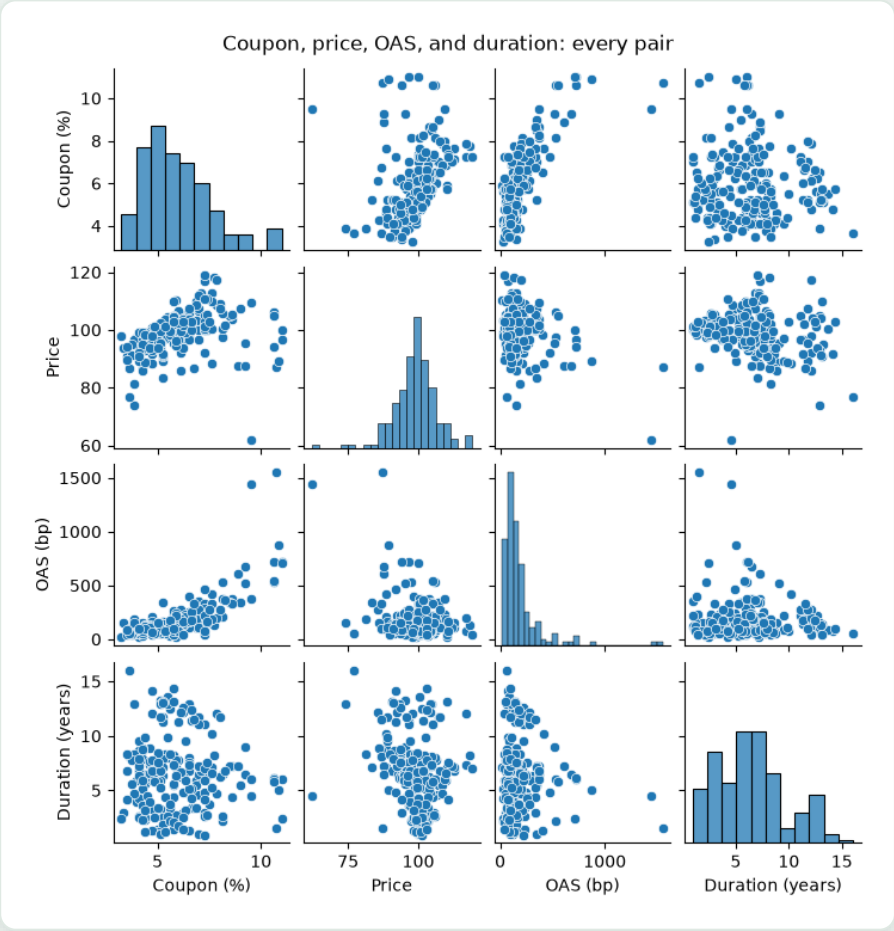
The same in Excel

Insert, Insert Scatter (X, Y) or Bubble Chart, Scatter

`=CORREL(M2:M201, L2:L201)` returns 0.99

Colour by group: one data series per grade

Pair plot: every pair at once




```
grid = sns.pairplot(pair_data, height=1.8)
```

What to read off it

16

panels: 4 rows and 4 columns

4

histograms, on the diagonal

6

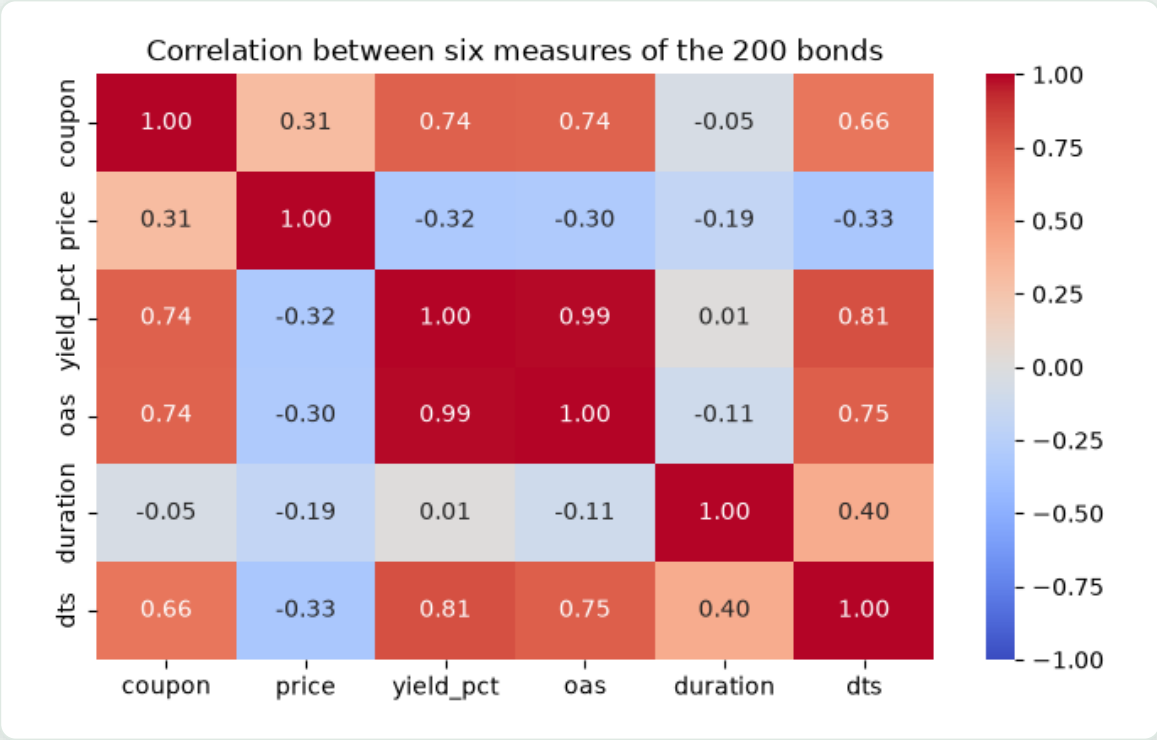
different pairs, each drawn twice

Read along a row. In the OAS row, the panel against coupon rises to the right, and the panel against duration is a flat cloud. The two widest bonds, at 1,446 and 1,551 bp, stand apart in every panel of that row.

The same in Excel

Excel has no single chart for this. The same picture is six separate Scatter charts, one for each pair of columns, arranged on a sheet.

Heatmap: a table with colour behind it




```
sns.heatmap(corr_table, annot=True, fmt='.2f',
            cmap='coolwarm', vmin=-1, vmax=1, ax=ax)
```

What to read off it

0.99

yield and OAS, the darkest

0.74

coupon and yield

-0.11

duration and OAS, a pale cell

Red is positive, blue is negative, pale is near zero. DTS shows both parents: 0.75 with OAS and 0.40 with duration. Counts take one colour: in sector by bucket, the darkest cell is Financials BBB, with 14 bonds.

The same in Excel

Home, Conditional Formatting, Color Scales

`=CORREL(M2:M201, N2:N201)` returns -0.11

Pivot table: sector in Rows, bucket in Columns

Choosing a chart

The question	Chart	matplotlib	seaborn	Excel chart
How is one numeric column distributed?	Histogram	`ax.hist()`	`sns.histplot()`	Histogram
How does that differ between groups?	Box plot	`ax.boxplot()`		Box and Whisker
How does one number compare by category?	Bar chart	`ax.bar(), ax.barh()`	`sns.barplot()`	Clustered Column or Bar
How does a number change along an order?	Line plot	`ax.plot()`		Line
Do two numeric columns move together?	Scatter plot	`ax.scatter()`	`sns.scatterplot()`	Scatter
What do all the pairs look like?	Pair plot		`sns.pairplot()`	several Scatter charts
Where are the large and small cells?	Heatmap		`sns.heatmap()`	Color Scales

The habits, whichever chart it is

Work out the numbers first, and check the chart against them.
Give every chart a title, and every axis a label with its unit.
Put ratings in credit order. No library does it for you.
Know how many bonds are behind each bar, box, or point on a line.
Describe what the chart shows. It is a picture of the data and not a view on any bond.

The AI check

Every notebook ends with code written in the style of an AI assistant. It has one bug. Find it.

1

Read

Read the code before you run it.

2

Predict

Decide what it should produce.

3

Run

Run it and look at the output.

4

Check

Compare with a number you worked out yourself.

Code that runs is not the same as code that is right.

What you can do now

- Make a figure and an axes, draw on it, and label every axis with its unit
- Show a distribution as a histogram, and compare groups with box plots
- Turn a grouped table into a bar chart or a line, in an order that makes sense
- Plot two measures against each other, with a correlation next to the picture
- Take a first look with a pair plot, and show a table as a heatmap

Week 3, Day 2 More Visualization: regression plots, violin plots, interactive charts, and customizing a chart